THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** erect and enlarged	Shaving mirror
2	**(c)** erect and diminished	Smaller view
3	**(d)** an erect, magnified image	Object within focus
4	**(c)** 20 minutes	A very small difference
5	**(b)** solstices	Around June 21 and December 21
6	**(d)** 11 days	365 − 354
7	**(d)** herbivores	Primary consumers
8	**(c)** monoculture	Reduces biodiversity
9	**(d)** saprotrophs	Feed on rotten matter
10	**(a)** the first cell formed after fertilisation	Develops into embryo
11	**(b)** keeps Earth warm by trapping heat	Natural, essential
12	**(d)** inside the female's body	Internal development

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Concave curves inwards and converges light (1). Convex bulges outward and diverges light (1).	1+1
14	22 March (1). 21 March in leap years (1).	1+1
15	Changing shape of the illuminated part seen from Earth (1). Occurs over nearly 29.5 days (1).	1+1
16	Sequence showing who eats whom (1). Grass → grasshopper → frog → snake (1).	1+1
17	Atmosphere – air (1). Hydrosphere – water, geosphere – solid Earth, biosphere – life (any one) (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Angle of incidence is 0° (1). Angle of reflection is 0° (1). Ray reflects back along the same path (1).	1+1+1
19	Earth takes slightly less than 365¼ days (1). Skipping century years corrects it (1). Years divisible by 400 are leap years (1).	1+1+1
20	Producers make food, e.g. plants (1). Consumers eat others, e.g. deer, lion (1). Decomposers break dead matter, e.g. fungi (1).	1+1+1
21	Fertilisation inside the female (1). Zygote enclosed in egg and laid (1). Embryo develops during hatching (1).	1+1+1
22	Gases trap heat (1). Keeps Earth warm (1). Without it Earth would be too cold (1).	1+1+1
	OR: Venus has dense carbon dioxide atmosphere (1). Strong greenhouse effect makes it very hot (1). Earth's effect keeps a moderate temperature (1).	1+1+1
23	Reflection: light bounces back; refraction: light passes with bending (1). Same medium vs different medium (1). Mirror vs lens (1).	1+1+1
	OR: Angle of reflection 25° (1). Reflected ray on opposite side of normal (1). Angle with mirror 65° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Launched by rockets (1). Communication (1). Earth observation (1). Astronomy (1). Examples Cartosat, AstroSat (1).	1+1+1+1+1
	OR: Divisible by 4 is leap (1). Century years skipped unless divisible by 400 (1). 1900 not a leap year (1). 2000 leap year (1). Reason: Earth takes slightly less than 365¼ days (1).	1+1+1+1+1
25	Pollination (1). Gametes (1). Fertilisation (1). Zygote to seed (1). Seed dispersal and germination (1).	1+1+1+1+1
	OR: Climate change (1). Pollution (1). Biodiversity loss (1). Individual actions (1). Community and global action (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** convex lens	Enlarges when close
2	**(b)** 0°	Retraces its path
3	**(c)** straight lines in a uniform medium	Rectilinear propagation
4	**(b)** sidereal year	Slightly longer than tropical year
5	**(d)** phases	Illuminated part changes
6	**(c)** full moon	Purnima
7	**(a)** the right conditions for an organism to live and grow	Home of an organism
8	**(d)** mangrove forests	At Ganga–Brahmaputra delta
9	**(b)** food web	More realistic picture
10	**(a)** pollination	By wind, insects, birds
11	**(b)** continuity of life	Species survive
12	**(b)** water	Oceans, rivers, ice

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	35° (1). 70° (1).	1+1
14	New moon (1). Full moon (1).	1+1
15	Network of interconnected food chains (1). Shows many feeding relationships in an ecosystem (1).	1+1
16	They break down dead plants and animals (1). Recycle nutrients to soil (1).	1+1
17	Atmosphere traps some of the Sun’s heat (1). Keeps Earth warm enough for life (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Concave reflectors direct sunlight to a point (1). Light converges onto a cooking pot (1). Heat is produced to cook food (1).	1+1+1
19	Uses Moon’s phases for months (1). Adds extra months to match the seasons (1). Traditional Indian calendars (1).	1+1+1
20	Ancient text on plant care (1). Stresses soil nourishment by organic manure (1). Kunapa Jala is a fermented liquid fertiliser (1).	1+1+1
21	Water stays liquid (1½). Temperature is neither too hot nor too cold (1½).	1½+1½
22	Keep a pencil in an empty tumbler and see it straight (1). Fill half with water (1). Pencil appears bent due to refraction (1).	1+1+1
	OR: Convex gives wide view (1½). Concave gives enlarged image (1½).	1½+1½
23	Noticed phases repeat every 29.5 days (1). Month defined by cycle (1). 12 months made the lunar year (1).	1+1+1
	OR: New moon to waxing crescent to first half (1). Gibbous to full moon (1). Waning back to new moon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Far away: inverted image (1). Image changes size as she moves closer (1). Close enough: erect and magnified (1). Distance decides image (1). Concave mirror converges (1).	1+1+1+1+1
	OR: Shape (1). Image type (1). Size (1). Lateral inversion (1). Use (1).	1+1+1+1+1
25	Mutualism (1½). Commensalism (1½). Parasitism (1). Examples (1).	1½+1½+1+1

	OR: Largest mangrove forest (1). Protects from storms (1). Biodiversity (1). World Heritage Site (1). Threats like pollution and sea-level rise (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** convex lens	Converging; real and inverted possible
2	**(a)** curved surface(s)	Bends light
3	**(a)** concave lens	Diverging lens for myopia
4	**(a)** new moon	Dark side faces us
5	**(c)** Vikram Sarabhai	Founder of ISRO's early efforts
6	**(d)** 21 March 1956	1 Chaitra 1878 Saka
7	**(d)** compete with native species	Not native
8	**(d)** conservation of ecosystems	Protected areas
9	**(d)** producers	They make food
10	**(a)** methane	Trap heat
11	**(b)** disturb climate patterns	Serious consequences
12	**(a)** distance from the Sun, atmosphere and water	Several conditions together

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Left side of object appears as right in image (1). Seen in all mirrors (1).	1+1
14	Lunar follows the Moon's phases (1). Solar follows the seasons (1).	1+1
15	Both organisms benefit (1). Bee and flower (1).	1+1
16	Region around a star where water stays liquid (1). Also called Goldilocks zone (1).	1+1
17	Holds the atmosphere (1). Not too strong to crush creatures (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Place a mirror upright on paper and draw normal (1). Send a ray with a pin or torch and mark the reflected ray (1). Measure angles; they are equal (1).	1+1+1
19	They follow the lunar calendar of 354 days (1½). So they shift about 11 days earlier every year (1½).	1½+1½
20	Mutualism: bee and flower (1). Commensalism: orchid on tree (1). Parasitism: tapeworm in humans (1).	1+1+1
21	Regrowth of a body from a small piece (1½). Planaria (1½).	1½+1½
22	Moon moves in orbit around Earth (1½). The lit part visible to us decreases (1½).	1½+1½
	OR: Cartosat – mapping (1). AstroSat – astronomy (1). AzaadiSat or InspireSat-1 – student or small satellite (1).	1+1+1
23	Organic manure (1). Crop rotation (1). Reducing synthetic chemicals (1).	1+1+1
	OR: Pests become resistant (1). Pollinators decrease (1). Biodiversity falls (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Solar sidereal festivals fixed, e.g. Pongal (2). Lunar festivals shift (2). Reason: lunar year 354 days (1).	2+2+1
	OR: Lunar: Moon's cycles, year 354 days (1½). Solar: seasons, 365 days with leap years (1½). Luni-solar: Moon months plus extra month (2).	1½+1½+2
25	Gases trap heat (1). Keeps Earth warm (1). Burning fossil fuels raises CO_2 (1). Methane from other sources (1). Global temperatures rise (1).	1+1+1+1+1
	OR: Fish and frogs fertilise in water (1½). Embryo develops in water (1). Birds fertilise inside, egg laid (1½). Mammals develop inside (1).	1½+1+1½+1

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